Task 1: Working AgentCore Harness with Knowledge Base Retrieval

System: Northstar Assist. **Account:** Udacity Cloud Lab (…4976), us-east-1. **Date:** 2026-10-10.

1. What was deployed

Resource	Name / ID	Key settings	Evidence
S3 bucket	`northstar-assist-kb-5ca661ad`	30 documents, public access blocked, SSE-S3, versioning on	`evidence/task1/s3_*.json`
Managed Knowledge Base	`northstar-assist-kb` (`WUWHZC06MP`)	Type MANAGED, Titan Text Embeddings V2 (`amazon.titan-embed-text-v2:0`), 1024 dimensions, FLOAT32	`evidence/task1/kb_details.jpg`, `knowledge_base.json`
Data source	`northstar-documents` (`DR2JEWLD2V`)	S3, managed parser, default chunking, on-demand sync	`evidence/task1/kb_sync.jpg`
AgentCore Gateway	`northstar-assist-gateway` (`northstar-assist-gateway-hd1uvsa8f3`)	MCP, inbound auth AWS_IAM	`evidence/task1/gateway.json`
Gateway target	`northstar-kb` (`5QFJRBLA4V`)	Connector `bedrock-knowledge-bases`, tool `Retrieve`, standard retrieval, 5 results	`evidence/task1/gateway_target_5QFJRBLA4V.json`
AgentCore Harness	`NorthstarAssist` (`NorthstarAssist-XSASd9gyWX`)	Claude Haiku 4.5 via US inference profile, course system prompt, memory disabled,	`evidence/task1/harness.json`

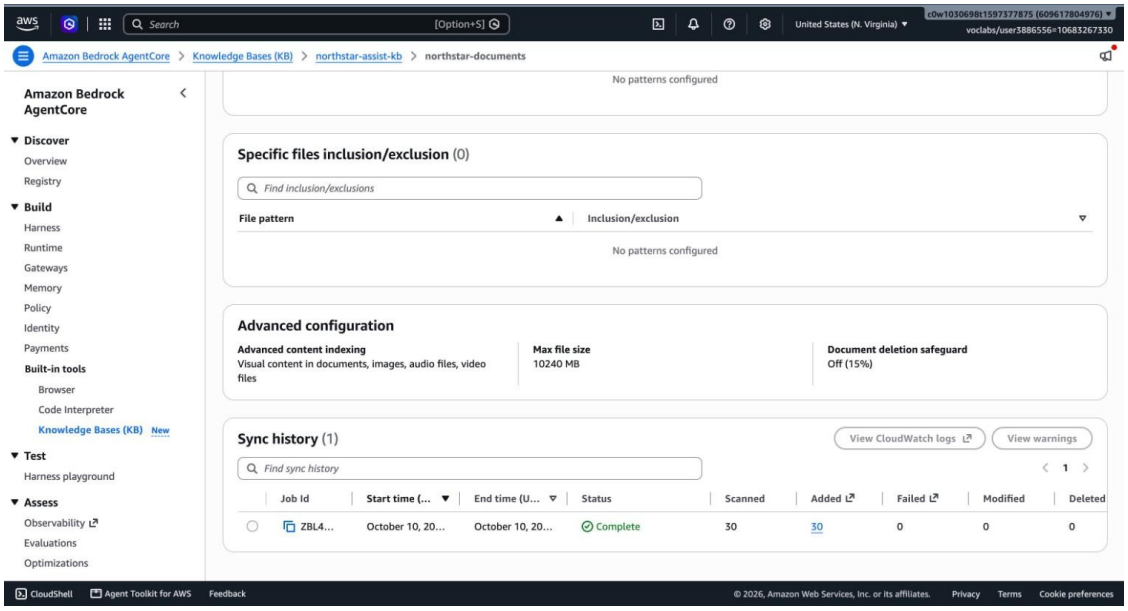
Resource	Name / ID	Key settings	Evidence
		gateway tool only	

Request path:

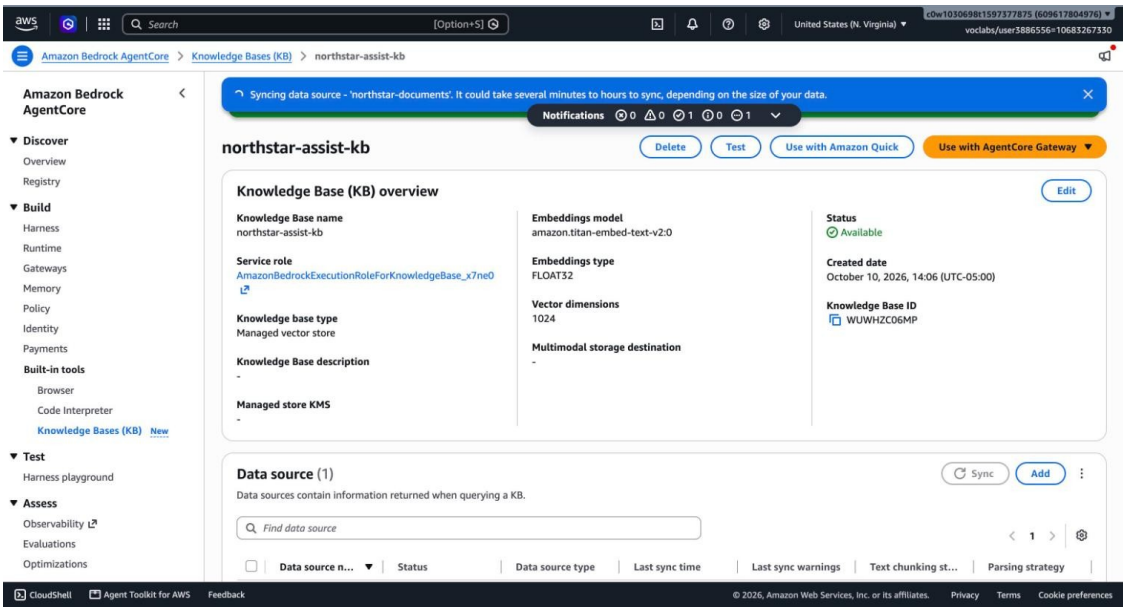
```
Employee → Harness (Claude Haiku 4.5) → Gateway tool → AgentCore Gateway → Managed KB → S3 documents
```

2. Knowledge base sync completed without errors

The data-source sync job `ZBL4LAMGLO` finished with status **COMPLETE**: **30 scanned, 30 indexed, 0 failed, 0 skipped**.



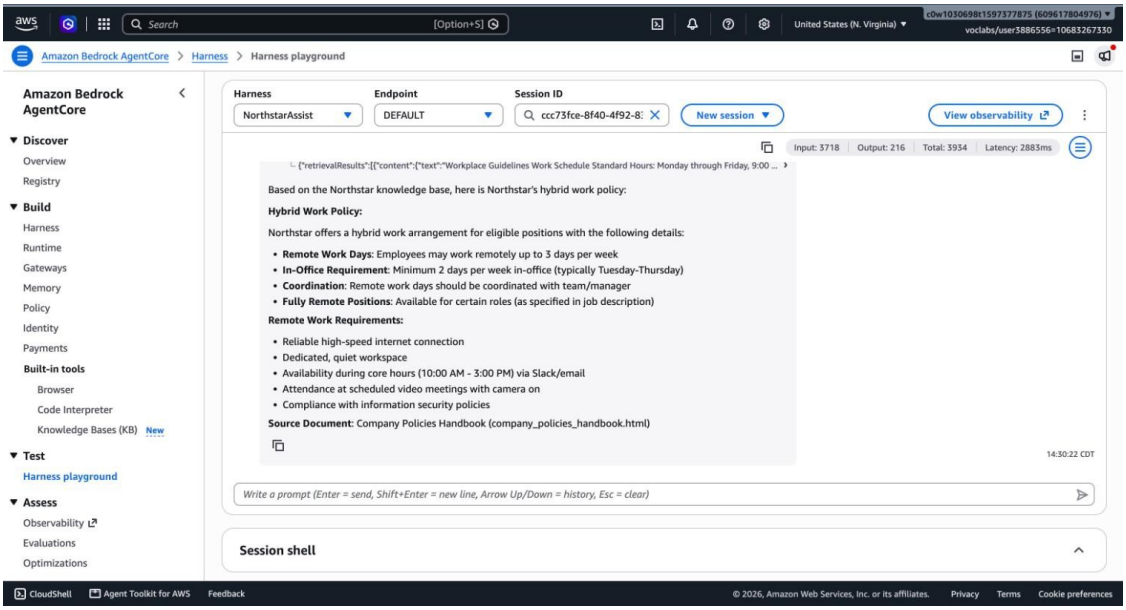
The same result from the API is saved in `evidence/task1/kb_ingestion_jobs.json`.



3. The harness answers a prompt using a foundation model

Prompt sent in the Harness playground and through the `InvokeHarness` API:

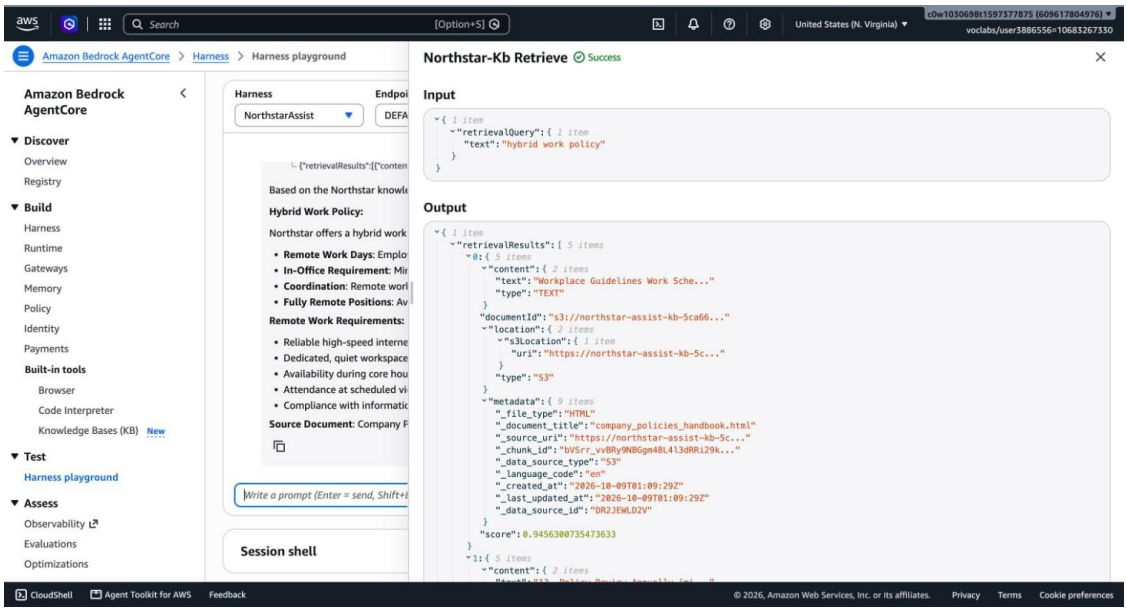
> What is Northstar's hybrid work policy? Name the source document.



Playground metrics for this turn: 3,718 input tokens, 216 output tokens, 2,883 ms.

4. The answer uses content retrieved from the knowledge base

The Agent trace shows the harness calling the **Northstar-Kb Retrieve** tool (status Success) before answering.



Trace field	Value
Tool	`northstar-kb___Retrieve`
Input	`{"retrievalQuery": {"text": "hybrid work policy"}}`
Results returned	5 chunks
Top result	`company_policies_handbook.html`, relevance score 0.946
Sources across the 5 chunks	`html/company_policies_handbook.html`, `docx/security_policy.docx`

Scripted transcript (InvokeHarness API)

Produced by `tools/run_suite.py --smoke`; the full record, including every retrieved chunk, is in `evidence/task1/smoke_baseline_20261010T192934Z.json`. Verdict: **ANSWERED_WITH_RETRIEVAL**, 5 chunks.

```
I'll search the Northstar knowledge base for information about the
hybrid work policy.Based on the knowledge base search, here is
Northstar's hybrid work policy:

**Hybrid Work Policy:**

Northstar offers a hybrid work arrangement for eligible positions with
the following guidelines:

- **Remote Work Allowance:** Employees may work remotely up to 3 days
per week
- **In-Office Requirement:** Minimum 2 days per week in-office
(typically Tuesday-Thursday)
```

```
- **Coordination:** Remote work days should be coordinated with
team/manager
- **Fully Remote Positions:** Available for certain roles (as
specified in job description)

**Remote Work Requirements:**
When working remotely, employees must have:
- Reliable high-speed internet connection
- Dedicated, quiet workspace
- Availability during core hours (10:00 AM - 3:00 PM) via Slack/email
- Attendance at scheduled video meetings with camera on
- Compliance with information security policies

**Source Document:** Company Policies Handbook
(company_policies_handbook.html)
```

The answer's facts (up to 3 remote days, minimum 2 in-office days, core hours 10:00–15:00) all appear in the first retrieved chunk, and the answer names the source document. This is the retrieved content in the model's context window, not model memory.

5. Notes for the later tasks

- **Console issue.** The console's Create Harness form created the default execution role and its policies, then stayed on "being provisioned" without creating the harness. The harness was then created with `aws bedrock-agentcore-control create-harness`, using that console-created role and the same settings. The role and its policies are therefore the unmodified console defaults that Task 4 reviews (`iam/before/`).
- **Baseline IAM is broad.** The default harness role allows model invocation on `foundation-model/*` and on every Bedrock resource in the account, plus Browser, Code Interpreter, EFS, S3 Files and Memory statements that this harness does not use.
- **Invocation-time overrides.** The playground and the `InvokeHarness` API accept a per-request model, system prompt and tool list. Whether a caller can use this to bypass the guardrail was tested and closed in Task 4 (`deliverables/04_iam_hardening.md`).
- **Chunks reach the model unfiltered.** Each retrieved chunk carries document text and metadata straight into the model's context. No guardrail is attached at this stage.
- **Runtime log group exists:** `/aws/bedrock-agentcore/runtimes/harness_NorthstarAssist-92Prr07JAK-DEFAULT`.